

Multivariate Lab-5**Date: 23-02-2026**

The data of weight (kg), height (cm) and chest circumference (cm) of 15 UG students selected from the multivariate normal distribution with mean vector $\mu'_0 = [58, 157, 42]$ and variance-

covariance matrix $\Sigma = \begin{bmatrix} 44 & 49 & 21 \\ 49 & 61 & 27 \\ 21 & 27 & 17 \end{bmatrix}$, are given below.

Weight(X_1)	Height(X_2)	Chest circumference(X_3)
59	152	35
63	157	46
57	156	40
47	147	33
66	166	47
68	168	42
70	171	45
52	151	38
56	154	44
53	152	41
56	156	42
64	169	48
62	163	42
48	151	39
51	153	41

Test the hypothesis $H_0 : \mu' = \mu'_0$ i.e. Mean vector of means of weight, height and chest circumference are taken from population with mean vector μ'_0 against $H_1 : \mu' \neq \mu'_0$ i.e. Mean vector of means of weight, height and chest circumference are not taken from population with mean vector μ'_0 at given level of significance $\alpha = 0.05$ [$\chi^2_{3, 0.05} = 7.81$].

Step 1 :- Hypothesis Statement :

Null Hypothesis : Means of (weight, height and Chest Circumference) are taken from the Population .

$$H_0 : \mu' = \mu'_0$$

Alternate Hypothesis : Means of (Weight , Height and Chest Circumference) are significantly differ from the Population .

$$H_1 : \mu' \neq \mu'_0$$

Step 2 :- Data Analysis :

```
mu_not<-matrix(data = c(58,157,42),nrow=3,ncol=1)

cov_matrix<-matrix(data = ,c(44,49,21,49,61,27,21,27,17),3,3)

weight<-c(59,63,57,47,66,68,70,52,56,53,56,64,62,48,51)
height<-c(152,157,156,147,166,168,171,151,154,152,156,169,163,151,153)
C_c<-c(35,46,40,33,47,42,45,38,44,41,42,48,42,39,41)

matrices<-matrix(data = c(weight,height,C_c),15,3)

x_bar<-colMeans(matrices)

inverse_cov<-solve(cov_matrix)

x_mu<-x_bar - mu_not

trnaspose<-t(x_mu)

result<-15%*trnaspose%*inverse_cov%*x_mu
result

##           [,1]
## [1,] 3.040571
```

Step 3 :- Decision :

So, $\chi^2_{cal} = 3.04 < \chi^2_{3, 0.05} = 7.81$ (at level of significance $\alpha = 0.05$) therefore, By according to given data we are fail to reject the Null Hypothesis.

Step 4 :- Interpretation :

The Means Vector of (Weight , Height and Chest Circumference) is equal to mean vector of the Population .